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EXAMINER	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 19/363,044	Applicant(s) Fang et al.	
	Examiner ADAM ALHARBI	Art Unit 3663	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/20/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10/20/2025 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

This Office Action is in response to the application filed on Oct. 20, 2025. Claims 1-20 are presently pending and are presented for examination.

Double Patenting

Claims 1-20 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 3, 11-23, 28, 30, and 35-47 of of U.S. Patent No. 12073664 as follows:

Current Application's claims	Parent Application's claims
1	1
2	1 & 28
3	21 & 45
4	20 & 44
5	20 & 44
6	20 & 44

7	3 & 30
8	11 & 35
9	12 & 36
10	13 & 37
11	14 & 38
12	15-16 & 39-40
13	15-16 & 39-40
14	17 & 41
15	18 & 42
16	19–20 & 43–44
17	19–20 & 43–44
18	21–23 & 45–47
19	21–23 & 45–47
20	21–23 & 45–47

Although the claims at issue are not identical, they are not patentably distinct from each other because current claims additionally include Interpreting a set of policies rather than a single policy. Claim 1 of the current application recites a policy acquisition circuit structured to interpret a "set of

data collection policies" and evaluate each policy in the set. In contrast, Claim 1 of the parent Application recites interpreting "a data collection policy". Multiplying or pluralizing a single operational element—specifically expanding a circuit from handling a single data collection policy to processing a set or plurality of policies—is a conventional engineering design choice. A person having ordinary skill in the art (POSITA) would consider executing multiple data collection policies in parallel to be a routine and obvious modification to optimize automotive parameter collection.

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq*. for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-2 and 4-20 are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Pub. No. 20170288886 (hereinafter, "Atarius") in view of U.S. Pat. No. 7917253 (hereinafter, "Inbarajan").

Regarding claim **1,** Atarius discloses **an apparatus, comprising:**

a policy acquisition circuit ("The policy and charging control (PCC) rules for the default EPS bearer may be pre-defined in the packet data network (PDN) gateway (GW)" (para 0167)) **structured to interpret a set of data collection policies each comprising:**

at least one requested vehicle property ("In an example embodiment, V2X UE may comprise capabilities for vehicle-to-vehicle service, vehicle-to-pedestrian service, vehicle-to-infrastructure service, and vehicle-to-network service capabilities" (para 0186))**, and**

a policy type that comprises an on demand policy ("In an example embodiment, a group of vehicle may periodically with periodicity of one message per second, broadcast/multicast the cooperative adaptive cruise control message (CACC) about the gap policy, speed, position in the CACC group to other vehicles" (para 0470))**;**

a policy processing circuit structured to, for each data collection policy in the set of data collection policies ("The MME may perform authentication for the V2X UE due to operator policy" (para

0248)), **determine a property request value in response to the at least one requested vehicle property** ("In an example embodiment, V2X application function (AF) may offer an application requiring dynamic policy and/or charging control. In an example embodiment, V2X AF may communicate with policy and charging rules function (PCRF) to transmit the dynamic session information" (para 0249));

a parameter acquisition circuit structured to, for each data collection policy in the set of data collection policies:

interpret at least one vehicle parameter value in response to the property request value and the policy type ("In an example embodiment, an emergency vehicle such as an ambulance, a police car broadcasts/multicasts periodically message CAM comprising its location, speed, and direction information to assist other vehicle to allow it a passage path. The periodicity may be adjusted by the MNO policy and/or other circumstances but may be limited to 10 per second" (para 0468)),

discontinue evaluating the data collection policy for data collection operations in response to fulfilling a data collection cycle of the data collection policy ("In an example embodiment, V2X application function (AF) may offer an application requiring dynamic policy and/or charging control. In an example embodiment, V2X AF may communicate with policy and charging rules function (PCRF) to transmit the dynamic session information. PCRF may use that information to control the V2X service data flow, authorization of V2X service on incomplete service information (which may be used in e.g. IMS session setup when terminating side needs to resource allocate), gating, quality of service (QoS), and/or flow based charging" (para 0249) and "if this is a new request, optional requested discovery timer set to validity timer associated with the expected V2X restricted code from the HPLMN V2X function (if it is set to zero, the V2X UE is requesting to remove the discovery entity ID and release the associated resources)" (para 0290)), **and**

delete deactivate the data collection policy in response to the parameter acquisition circuit discontinuing the evaluating the data collection policy ("if this is a new request, optional requested

discovery timer set to validity timer associated with the expected V2X restricted code from the HPLMN V2X function (if it is set to zero, the V2X UE is requesting to remove the discovery entity ID and release the associated resources)" (para 0290))**; and**

However, Atarius does not explicitly teach

a parameter provisioning circuit structured to, for each data collection policy in the set of data collection policies, selectively transmit the at least one vehicle parameter value in response to the data collection policy.

Inbarajan, in the same field of endeavor, teaches

a parameter provisioning circuit structured to, for each data collection policy in the set of data collection policies, selectively transmit the at least one vehicle parameter value in response to the data collection policy ("telematics unit 30 receives various types of vehicle-related data from one or more modules, units, components, devices, programs, etc. operating throughout the vehicle, step 104. Examples of acceptable vehicle-related data include: a vehicle mileage reading, a vehicle location (past and/or present), a vehicle history (record of driving times, average driving speed, etc.), a diagnostic trouble code (DTC), an engine oil life reading, emissions data, tire pressure, and a vehicle identification number (VIN), to name but a few possibilities. It should be appreciated that the aforementioned examples are only some of the types of vehicle-related data that can be gathered by telematics unit 30, as other types of information relating to the vehicle could be used as well" (Col. 6, lines 10-30))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to wirelessly transmits the vehicle-related data from the telematics unit to call center 20 over wireless carrier system; see Inbarajan at least at [Col. 6, lines 10-30].

Regarding claim **2,** Atarius discloses **the apparatus of claim 1.** Additionally, Atarius discloses **wherein the parameter acquisition circuit is further structured to discontinue evaluating the data**

collection policy for data collection operations prior to fulfillment of the data collection cycle in response to receiving a cancellation instruction ("if this is a new request, optional requested discovery timer set to validity timer associated with the expected V2X restricted code from the HPLMN V2X function (if it is set to zero, the V2X UE is requesting to remove the discovery entity ID and release the associated resources)" (para 0290))**.**

Regarding claim **4,** Atarius discloses **the apparatus of claim 1.** Additionally, Atarius discloses **wherein the parameter acquisition circuit is further structured to initiate data collection operations only in response to conditions defined within the data collection policy** ("a vehicle in case of emergency stop may broadcast/multicast periodically a message DENM with the stationary vehicle warning to alert the other cars about its current situation. The periodicity may be adjusted by the MNO policy and/or other circumstances but may be limited to 10 per second. The size of the broadcasted/multicasted DENM may be 400 Bytes but with the maximum of 1200 Bytes and should have a latency less than 100 msec. Both E-UTRAN or direct communication may support communication range to give the UEs ample response time e.g. 4 seconds" (para 0469))**.**

Regarding claim **5,** Atarius discloses **the apparatus of claim 1.** However, Atarius does not explicitly teach **wherein at least one data collection policy specifies a finite number of data collection events to be executed, after which the data collection policy is deactivated.**

Inbarajan, in the same field of endeavor, teaches

wherein at least one data collection policy specifies a finite number of data collection events to be executed, after which the data collection policy is deactivated ("certain portions of the vehicle-related data are deleted or otherwise removed from the third party database in response to one of a number of different events, step 114. For example, each piece of data saved to the third party database could be given an expiration so that when the expiration occurs, that data is deleted or otherwise removed. In this embodiment, the authorized third party is provided with a window of time in which

they are able to retrieve the designated data. After the expiration of this time period, the designated data is automatically deleted from the third party database and is no longer available for retrieval" (Col. 9, Lines 28-44)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to set the data in the third party database to expire on the same day of each month; see Inbarajan at least at [Col. 9, Lines 28-44].

Regarding claim **6,** Atarius discloses **the apparatus of claim 5.** However, Atarius does not explicitly teach **further comprising deactivating the data collection policy after the finite number of data collection events.**

Inbarajan, in the same field of endeavor, teaches

further comprising deactivating the data collection policy after the finite number of data collection events ("According to a different embodiment, the third party database can be set up such that certain vehicle-related data is deleted after an authorized third party has accessed the information; that is, it is only available for one retrieval" (Col. 9, Lines 45-55)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to remove only that data that has been accessed by the third party; see Inbarajan at least at [Col. 9, Lines 60-65].

Regarding claim **7,** Atarius discloses **the apparatus of claim 1.** However, Atarius does not explicitly teach **wherein the policy acquisition circuit is further structured to mark for deactivation delete the data collection policy in response to the parameter provisioning circuit transmitting the at least one vehicle parameter value corresponding to the data collection policy.**

Inbarajan, in the same field of endeavor, teaches

wherein the policy acquisition circuit is further structured to mark for deactivation delete the data collection policy in response to the parameter provisioning circuit transmitting the at least one

vehicle parameter value corresponding to the data collection policy ("certain portions of the vehicle-related data are deleted or otherwise removed from the third party database in response to one of a number of different events, step 114. For example, each piece of data saved to the third party database could be given an expiration so that when the expiration occurs, that data is deleted or otherwise removed. In this embodiment, the authorized third party is provided with a window of time in which they are able to retrieve the designated data. After the expiration of this time period, the designated data is automatically deleted from the third party database and is no longer available for retrieval" (Col. 9, Lines 28-44))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to set the data in the third party database to expire on the same day of each month; see Inbarajan at least at [Col. 9, Lines 28-44].

Regarding claim **8,** Atarius discloses **the apparatus of claim 1.** However, Atarius does not explicitly teach **wherein the data collection policy further comprises a transmission description value corresponding to the at least one requested vehicle property.**

Inbarajan, in the same field of endeavor, teaches

wherein the data collection policy further comprises a transmission description value corresponding to the at least one requested vehicle property ("the electronic tag can also indicate to which third parties the data will be made available. According to different embodiments, the third party database can be organized such that various files of vehicle-related data are grouped together within the database and are given a tag for the entire group, or it can be organized such that each vehicle-related data file is individually tagged" (Col. 8, Lines 28-37))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to make an electronic tag, which can be in the form of a flag; see Inbarajan at least at [Col. 8, Lines 28-37].

Regarding claim **9,** Atarius discloses **the apparatus of claim 8.** Additionally, Atarius discloses **wherein the transmission description value comprises at least one value selected from the values consisting of:**

a transmission priority value ("The high priorities of UE's V2X capabilities may be assigned to law enforcement, medical units, public safety units, etc. In an example embodiment, V2X services may be V2V service, V2P service, V2I service, and/or V2N service. The information element comprising the priorities may apply to all V2X services" (para 0169))**;**

a data storage description ("The base station 401 may include at least one communication interface 402, at least one processor 403, and at least one set of program code instructions 405 stored in non-transitory memory" (para 0152))**;**

a network zone utilization description ("The MME may send GTP: session management response message comprising EPS bearer ID and the V2X user location towards the PDN-GW. The PDN-GW may send a provision acknowledge message towards PCRF to indicate that the QoS2 could have been enforced and it may inform the PCRF about user location information and the V2X UE time zone" (para 0254))**; or**

an access point name (APN) value (" If the UE provides access point network (APN), the MME may use PDN-GW corresponding to that APN. If the UE does not provide APN, the MME may use PDN-GW corresponding to default APN provided by the HSS" (para 0182))**.**

Regarding claim **10,** Atarius discloses **the apparatus of claim 1.** However, Atarius does not explicitly teach **wherein the data collection policy further comprises a data configuration value, and wherein the policy processing circuit is further structured to determine the property request value in response to the data configuration value.**

Inbarajan, in the same field of endeavor, teaches

wherein the data collection policy further comprises a data configuration value, and wherein the policy processing circuit is further structured to determine the property request value in response to the data configuration value ("The priority level of the event can determine when the vehicle-related data associated with that event is sent to the call center. For example, vehicle-related data that is considered critical, such as safety-related data, could be assigned a high priority level and be sent as soon as it is detected by telematics unit 30 instead of waiting for a scheduled data transmission. Call center 20 would then be in a position to quickly disseminate the critical information to authorized third parties as soon as it was received. According to another embodiment, the vehicle-related data transmission of step 106 could be in response to a request initiated at call center 20 or at a third party and communicated directly to the vehicle or through the call center" (Col. 6, Lines 44-55))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to wirelessly transmits the vehicle-related data from the telematics unit to call center 20 over wireless carrier system; see Inbarajan at least at [Col. 6, lines 10-30].

Regarding claim **11,** Atarius discloses **the apparatus of claim 1.** Additionally, Atarius discloses **wherein the data collection policy further comprises a triggered data description** ("The request may be triggered directly by the AF or a request directed to the AF. The latter is shown in this example embodiment. In FIG. 21 example embodiment, the AF is triggered" (para 0254))**.**

Regarding claim **12,** Atarius discloses **the apparatus of claim 1.** Additionally, Atarius discloses **wherein the data collection policy further comprises a policy priority value** ("The high priorities of UE's V2X capabilities may be assigned to law enforcement, medical units, public safety units, etc. In an example embodiment, V2X services may be V2V service, V2P service, V2I service, and/or V2N service. The information element comprising the priorities may apply to all V2X services" (para 0169))**.**

Regarding claim **13,** Atarius discloses **the apparatus of claim 12.** Additionally, Atarius discloses **wherein the policy priority value comprises at least one priority value selected from the values consisting of:**

a data collection priority value ("the V2X subscriber may define the priority of the V2X communication prior to initiation of such communications. The V2X subscriber may inform the V2X AS by using the V2X UE about the priority of the V2X service. If the V2X service may be an IMS application, an authorized Resource-Priority header field comprising an appropriate namespace and priority value may be used in the SIP message sent by the V2X UE to the V2X AS to identify the priority of the V2X service" (para 0251))**;**

a data storage priority value ("The V2X UE may store QoS2, radio priority, packet flow ID from the session management request message and modified APN-AMBR" (para 0254))**; or**

a transmission priority value ("In an example, the embodiment may define a new dedicated AVP for V2X service, in the diameter AA-request transmitted by the V2X AF towards the PCRF, wherein the new AVP may modify the priority of the V2X service in terms of new QoS assignment" (para 0259))**.**

Regarding claim **14,** Atarius discloses **the apparatus of claim 1.** Additionally, Atarius discloses **wherein the policy acquisition circuit is further structured to determine a policy capability value, and to selectively enable the data collection policy in response to the policy capability value** ("the embodiment may register one or more capabilities such as for being V2X UE capable in the wireless network at the time of NAS attachment procedure. In an example, the embodiment may register the priority of one or more capabilities such as for being V2X UE capable in the wireless network at the time of NAS attachment procedure. In an example, the embodiment may register one or more capabilities such as for being V2X UE capable in the wireless network at the time of NAS attachment procedure by using NAS attach request message" (para 0175))**.**

Regarding claim **15,** Atarius discloses **the apparatus of claim 1.** Additionally, Atarius discloses **wherein the policy acquisition circuit is further structured to determine a policy authorization value, and to selectively enable the data collection policy in response to the policy authorization value** ("V2X application function (AF) may offer an application requiring dynamic policy and/or charging control. In an example embodiment, V2X AF may communicate with policy and charging rules function (PCRF) to transmit the dynamic session information. PCRF may use that information to control the V2X service data flow, authorization of V2X service on incomplete service information" (para 0249))**.**

Regarding claim **16,** Atarius discloses **the apparatus of claim 1.** Additionally, Atarius discloses **wherein the data collection policy comprises a policy life cycle description, and wherein the policy acquisition circuit is further structured to selectively enable the data collection policy in response to the policy life cycle description** ("The HPLMN V2X function based on the V2X application code (or V2X application code prefix with V2X application code suffix pool if restricted direct discovery with application-controlled extension was requested by the announcing V2X UE) and the corresponding validity timer, may allocate a discovery filter with corresponding time-to-live (TTL)" (para 0309))**.**

Regarding claim **17,** Atarius discloses **the apparatus of claim 16.** Additionally, Atarius discloses **wherein the policy life cycle description comprises at least one description selected from the descriptions consisting of:**

a policy start time ("V2X message start time which may indicate the time of the start of the V2X message; V2X finish time (absolute time referring when the transmission of the message ends" (para 0494))**;**

a policy end time ("V2X message start time which may indicate the time of the start of the V2X message; V2X finish time (absolute time referring when the transmission of the message ends" (para 0494))**;**

a triggered data description comprising a collection criteria value for the data collection policy ("The request may be triggered directly by the AF or a request directed to the AF. The latter is shown in this example embodiment. In FIG. 21 example embodiment, the AF is triggered" (para 0254));

an amount of data to be captured under the data collection policy ("Example criteria may be based, at least in part, on for example, traffic load, initial system set up, packet sizes, traffic characteristics" (para 0157));

a number of data collection events to be captured under the data collection policy ("In an example embodiment, a vehicle may broadcast/multicast periodically its location, speed, acceleration, and estimated trajectory to other vehicles to avoid or mitigate any collision. In an example embodiment, a vehicle may broadcast/multicast periodically its location, speed, acceleration, and estimated trajectory at the time of control loss to other vehicles to avoid or mitigate any collision. The periodicity may be adjusted due to MNO policy and/or other circumstances but may be limited to 10 per second. The size of the broadcasted/multicasted messages may be 50-300 bytes but may be up 1200 bytes" (para 0467));

or

a number of trigger events wherein data is to be captured under the data collection policy ("an emergency vehicle such as an ambulance, a police car broadcasts/multicasts periodically message CAM comprising its location, speed, and direction information to assist other vehicle to allow it a passage path. The periodicity may be adjusted by the MNO policy and/or other circumstances but may be limited to 10 per second. The size of the broadcasted/multicasted CAM may be 50-300 Bytes and should have a latency less than 100 msec" (para 0468)).

Regarding claim **18,** Atarius discloses **the apparatus of claim 16.** However, Atarius does not explicitly teach **wherein the policy life cycle description comprises:**

a past data recovery triggered in response to an event; or

an indicator of a buffer for storing historical data.

Inbarajan, in the same field of endeavor, teaches

a past data recovery triggered in response to an event ("As used herein, "temporary database" means a database in which individual items of customer-related data are maintained on a temporary basis; e.g., until a period of time has passed or some event (e.g., oil change, the data is accessed, etc.) has occurred" (Col. 5, Lines 31-50))**; or**

an indicator of a buffer for storing historical data ("The customer database can maintain a history of all the vehicle-related data associated with the customer and can allow the call center to provide a myriad of services to the vehicle and to the authorized user" (Col. 7, Lines 22-30))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to allow the call center to provide a myriad of services to the vehicle and to the authorized user and provide the authorized user with notification recommending that they seek appropriate maintenance to remedy the problem; see Inbarajan at least at [Col. 7, Lines 22-30].

Regarding claim **19,** Atarius discloses **the apparatus of claim 18.** Additionally, Atarius discloses **wherein the event is at least one of:**

an accident ("a vehicle may broadcast/multicast periodically its location, speed, acceleration, and estimated trajectory to other vehicles to avoid or mitigate any collision. In an example embodiment, a vehicle may broadcast/multicast periodically its location, speed, acceleration, and estimated trajectory at the time of control loss to other vehicles to avoid or mitigate any collision" (para 0467))**; or**

a component failure ("the V2X UE may comprise capabilities for the emergency vehicle-to-vehicle service, emergency vehicle-to-pedestrian service, emergency vehicle-to-infrastructure service, and emergency vehicle-to-network service capabilities" (para 0187))**.**

Regarding claim **20,** Atarius discloses **the apparatus of claim 18.** However, Atarius does not explicitly teach **wherein the buffer is a rolling buffer and the past data recovery retrieves the historical data stored in the rolling buffer.**

Inbarajan, in the same field of endeavor, teaches

wherein the buffer is a rolling buffer and the past data recovery retrieves the historical data stored in the rolling buffer ("The customer database is generally a permanent or semi-permanent database that is designed to store vehicle-related data such as DTCs, emissions data, vehicle mileage, tire pressure, oil life, geographic information, as well as any other pertinent vehicle-related data that is sent from the vehicle. In addition, the customer database could store subscriber authentication information, profile records, behavioral patterns, and other pertinent customer-related data. The third party database, on the other hand, is generally a temporary database that is coupled to the customer database and land network 16 so that it can exchange information with the customer database and various third parties" (Col. 5, Lines 31-50))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Inbarajan in order to provide a "temporary database" means a database in which individual items of customer-related data are maintained on a temporary basis; e.g., until a period of time has passed or some event (e.g., oil change, the data is accessed, etc.) has occurred; see Inbarajan at least at [Col. 5, Lines 31-50].

Claim 3 is rejected under 35 U.S.C. 103 as being unpatentable over U.S. Pub. No. 20170288886 (hereinafter, "Atarius"), in view of U.S. Pat. No. 7917253 (hereinafter, "Inbarajan"), as applied to claim 1 above, and in further view of U.S. Pub. No. 20110196571 (hereinafter, "Foladare").

Regarding claim **3,** Atarius discloses **the apparatus of claim 1.** However, Atarius does not explicitly teach **wherein the parameter acquisition circuit is further structured to recover and resume evaluation of a data collection policy following an interruption due to a shutdown event.**

Foladare, in the same field of endeavor, teaches

wherein the parameter acquisition circuit is further structured to recover and resume evaluation of a data collection policy following an interruption due to a shutdown event ("The data may be uploaded intermittently, either upon start-up of the engine, shut down of the engine, at specific time intervals, at specific mileage intervals, at service shops, or on demand or internally or externally generated request" (para 0020))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Atarius with the teachings of Foladare in order to provide a variable rate insurance plan based on real-time driving behavior; see Foladare at least at [0020].

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ADAM ALHARBI whose telephone number is (313)446-6621. The examiner can normally be reached on M-F 11:00AM – 7:30PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Abby Flynn can be reached on (571) 272-9855. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ADAM M ALHARBI/
Primary Examiner, Art Unit 3663